

Resolution Geometry of Coherence and Operational Time

Operator Duration, BCH Composition Defects, Scalar Characters, Universal Clock Factorization, and Spacetime No-Go Results

Matěj Rada

MatejRada@email.cz

Licensed under CC BY-NC-ND 4.0

August 2026

Abstract

Resolution Geometry (RG) begins from a declared experiment and its observational quotient, not from a spacetime manifold or a primitive scalar time. This paper characterizes the temporal structure that can be constructed from composable observable coherence processes. For a strictly positive self-adjoint contraction semigroup $C_t = e^{-tK}$, the canonical duration object is operator-valued:

$$\mathcal{T}_t = -\log C_t = tK.$$

It is natural under admissible transport and additive under semigroup composition. For ordered noncommuting positive processes, the failure of naive addition is the Baker–Campbell–Hausdorff (BCH) defect. Its leading term is the commutator. We carefully separate this algebraic composition defect from operational non-Markovian memory: noncommutativity alone is not a criterion for non-Markovianity. For scalar clocks, we prove that every continuous, conjugation-invariant, extensive multiplicative character on finite-dimensional positive-definite operators is a determinant power; hence the associated extensive volume clock is $a \operatorname{tr} \mathcal{T} = -a \log \det C$, unique up to a positive unit. Modal, extremal, and intensive clocks remain distinct declared scalarizations and are not falsely identified with this character.

A universal operational clock is characterized independently as a one-dimensional positive affine object through which compatible clock mechanisms factor. Positive affine overlap maps satisfying identity and cocycle laws yield a common clock unique up to global positive affine gauge; nonlinear or history-dependent recalibration is an obstruction. We then prove no-go results: unsigned duration does not determine time orientation; rank-deficient interval experiments do not identify a Lorentzian metric; observable Lorentzian rank does not identify latent dimension; and distinct latent dynamical realizations can induce the same complete observable path law. Consequently, RG may identify a four-dimensional Lorentzian observable quotient without asserting a uniquely four-dimensional spacetime ontology.

The complete hierarchy is therefore

$$\begin{aligned} (\mathcal{E}, \Gamma) &\rightarrow Q_{\mathcal{E}} \rightarrow \text{observable geometry} \rightarrow C \rightarrow \mathcal{T} = -\log C \\ &\rightarrow \text{scalar clock} \rightarrow \text{universal clock} \rightarrow \text{physical realization.} \end{aligned}$$

The paper states explicit hypotheses, theorem and corollary ladders, falsification criteria, reproducibility artifacts, and the honest remaining frontier: extension from positive operator semigroups to completely positive process tensors and experimental identification with relativistic proper time.

Contents

1 Introduction

2

2	RG kinematics and temporal input	3
3	Canonical operator duration	4
4	Ordered composition and the BCH defect	4
5	Scalar clocks and the trace character	5
6	Universal operational clock factorization	6
7	From coherence duration to physical proper time	7
8	No-go theorems for time orientation and metric recovery	8
9	Spacetime dimensionality and latent non-identifiability	8
10	Synthesis with fixed-protocol Resolution Geometry	9
11	Axiom independence and attack ledger	9
12	Non-claims and falsification criteria	10
13	Reproducibility	10
14	Open frontier	11
15	Conclusion	11
A	Finite attack summary	12
B	Notation ledger	12

1 Introduction

The question addressed here is not whether time can be declared equal to coherence. That claim would conflate an abstract compositional construction with a physical realization. The mathematically precise question is:

What additive object is canonically associated with observable coherence composition, how do admissible scalar clocks factor through it, and what can or cannot be inferred about physical time and spacetime from the resulting experiment?

RG supplies the required order of construction. A declared experiment $\mathcal{E}$ first determines the equality-of-law quotient $Q_{\mathcal{E}}$. A protocol Γ then supplies comparison, resolution, admissibility, and transport structure. Temporal structure enters only after a family of observable processes has been declared on that geometry. Thus spacetime is not a prerequisite of the abstract construction.

The central result is that the operator logarithm is the additive linearization of a strictly positive coherence semigroup. Scalar clocks arise only after an additional scalarization. Cross-mechanism physical clock universality is a separate factorization theorem, and relativistic proper time is a further empirical realization. This layered formulation prevents three common overclaims:

1. that an operator logarithm is automatically a unique scalar clock;

2. that noncommutativity is automatically operational memory;
3. that reconstruction of observable Lorentzian rank identifies latent spacetime ontology.

Contributions. We provide:

1. a precise domain for operator duration, including extinguished modes;
2. a natural additive operator-duration theorem;
3. the exact BCH composition defect for ordered noncommuting processes;
4. a no-go theorem separating BCH defect from non-Markovian memory;
5. a determinant-character theorem for extensive scalar clocks;
6. a universal affine-clock factorization theorem;
7. no-go theorems for orientation, metric reconstruction, latent dimension, and latent realization;
8. a synthesis with the fixed-protocol RG representation theorem;
9. explicit non-claims, falsification conditions, and reproducibility statements.

2 RG kinematics and temporal input

Definition 2.1 (Declared admissible experiment). A declared admissible experiment is a pair

$$(\mathcal{E}, \Gamma), \quad \mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta}),$$

where P_θ is the complete declared observation law and Γ is the admissibility protocol.

Definition 2.2 (Observational quotient). Define

$$\theta \sim_{\mathcal{E}} \theta' \iff P_\theta = P_{\theta'}.$$

The observational quotient is $Q_{\mathcal{E}} = \Theta / \sim_{\mathcal{E}}$.

Theorem 2.3 (Experiment-internal factorization). *Every experiment-internal map constant on equality-of-law classes factors uniquely through $Q_{\mathcal{E}}$.*

Proof. For an extensional map $f : \Theta \rightarrow Z$, define $\tilde{f}([\theta]) = f(\theta)$. Equality of laws makes the definition independent of representative. Surjectivity of the quotient map gives uniqueness. $\square$

Definition 2.4 (Observable coherence process). Let W be a regular finite-dimensional observable sector. A coherence process on W is a family $(C_t)_{t \geq 0}$ of bounded positive operators representing survival or contraction of observable coherence. In the basic semigroup regime we assume

$$C_0 = I, \quad C_{t+s} = C_t C_s, \quad 0 \prec C_t \preceq I,$$

with strong continuity in t .

Remark 2.5. Strict positivity for finite t ensures that the principal logarithm exists. If a mode reaches exact zero, duration is treated either on the support of C_t or as the extended value $+\infty$ on the extinguished mode.

3 Canonical operator duration

Theorem 3.1 (Generator and operator duration). *Let $(C_t)_{t \geq 0}$ be a strongly continuous self-adjoint contraction semigroup on a finite-dimensional Hilbert space. Then there is a unique positive-semidefinite generator K such that*

$$C_t = e^{-tK}.$$

The operator-valued duration

$$\boxed{\mathcal{T}_t = -\log C_t = tK}$$

is additive:

$$\mathcal{T}_{t+s} = \mathcal{T}_t + \mathcal{T}_s.$$

Proof. Finite-dimensional strong continuity gives an exponential representation $C_t = e^{tA}$. Self-adjoint contractivity implies $A = -K$ with $K \succeq 0$. Strict positivity gives the principal logarithm, so $-\log C_t = tK$. Additivity follows immediately. $\square$

Corollary 3.2 (Naturality). *If an admissible transport U conjugates the process,*

$$C'_t = UC_tU^{-1},$$

then

$$\mathcal{T}'_t = U\mathcal{T}_tU^{-1}.$$

Proof. The principal functional calculus is equivariant under similarity on the stated spectral domain. $\square$

Corollary 3.3 (Uniqueness of logarithmic linearization on scalar survival). *Let $h : (0, 1] \rightarrow [0, \infty)$ be continuous, strictly decreasing, $h(1) = 0$, and satisfy*

$$h(xy) = h(x) + h(y).$$

Then $h(x) = -a \log x$ for a unique $a > 0$.

Proof. Set $x = e^{-u}$ and $f(u) = h(e^{-u})$. Then $f(u+v) = f(u) + f(v)$. Continuity gives $f(u) = au$; strict decrease gives $a > 0$. $\square$

Non-claim 3.4 (No unique scalar clock yet). The canonical object $\mathcal{T}_t$ is operator-valued. A scalar clock requires a declared or characterized scalar functional. The theorem does not identify one spectral mode, determinant volume, worst-case mode, or mean mode without further operational assumptions.

4 Ordered composition and the BCH defect

Definition 4.1 (Ordered composition defect). Let

$$C_{01} = e^{-\mathcal{T}_{01}}, \quad C_{12} = e^{-\mathcal{T}_{12}}, \quad C_{02} = C_{12}C_{01},$$

and suppose a consistent logarithm branch is defined. Set

$$\mathcal{M}_{012}^{\text{BCH}} := \mathcal{T}_{02} - \mathcal{T}_{01} - \mathcal{T}_{12}, \quad \mathcal{T}_{02} := -\log C_{02}.$$

Theorem 4.2 (BCH composition defect). *In a convergent BCH neighbourhood,*

$$\mathcal{M}_{012}^{\text{BCH}} = \frac{1}{2}[\mathcal{T}_{01}, \mathcal{T}_{12}] + \frac{1}{12}([\mathcal{T}_{01}, [\mathcal{T}_{01}, \mathcal{T}_{12}]] + [\mathcal{T}_{12}, [\mathcal{T}_{12}, \mathcal{T}_{01}]] + \dots.$$

In particular, commuting durations compose additively.

Proof. Apply

$$\log(e^X e^Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}([X, [X, Y]] + [Y, [Y, X]]) + \dots$$

with $X = -\mathcal{T}_{12}$ and $Y = -\mathcal{T}_{01}$, then negate the resulting logarithm. $\square$

Corollary 4.3 (Leading noncommutative obstruction). *For $\mathcal{T}_{01} = \varepsilon A$ and $\mathcal{T}_{12} = \varepsilon B$,*

$$\mathcal{M}_{012}^{\text{BCH}} = \frac{\varepsilon^2}{2}[A, B] + O(\varepsilon^3).$$

Remark 4.4 (Global and positivity caveats). The BCH series is local unless stronger algebraic hypotheses hold. Moreover, the ordered product of noncommuting positive-definite self-adjoint operators need not be self-adjoint, although it has positive real spectrum under standard similarity arguments. If self-adjoint composite duration is required, one may instead use a symmetric product such as $C_{12}^{1/2} C_{01} C_{12}^{1/2}$, whose expansion begins at nested-commutator order. These are distinct composition protocols and must not be conflated.

No-go theorem 4.5 (BCH defect is not equivalent to non-Markovian memory). *Nonzero $\mathcal{M}_{012}^{\text{BCH}}$ does not by itself imply operational non-Markovianity, and vanishing $\mathcal{M}_{012}^{\text{BCH}}$ does not by itself prove operational Markovianity.*

Proof. A time-inhomogeneous but memoryless evolution can have noncommuting generators on successive intervals, producing a nonzero BCH defect while remaining divisible by process maps that depend only on the current state. Conversely, an operational process with hidden memory can induce commuting or even scalar reduced survival factors on a restricted observable sector, giving zero BCH defect there while multitime intervention statistics remain non-Markovian. Hence the BCH defect measures noncommutative logarithmic composition; operational memory requires a multitime process criterion. $\square$

Definition 4.6 (Operational memory obstruction). For scalar or commuting survival factors define

$$\mathcal{M}_{012}^{\text{op}} := \mathcal{T}_{02} - \mathcal{T}_{01} - \mathcal{T}_{12}.$$

More generally, an operational memory obstruction is any invariant that vanishes exactly when the declared multitime experiment factorizes according to the chosen Markov condition.

Proposition 4.7 (Cocycle consistency). *If a memory-corrected duration law*

$$\mathcal{T}_{02} = \mathcal{T}_{01} + \mathcal{T}_{12} + \mathcal{M}_{012}^{\text{op}}$$

is compatible with associative concatenation, then the memory terms satisfy the corresponding two-cocycle identity on triple compositions.

Proof. Compute the total duration by the two groupings $(C_{23}C_{12})C_{01}$ and $C_{23}(C_{12}C_{01})$. Equality of the total composite gives equality of the two sums of local durations and memory corrections, which is the cocycle equation. $\square$

5 Scalar clocks and the trace character

Definition 5.1 (Extensive scalar coherence character). A family of maps $\chi_n : \text{PD}_n(\mathbb{C}) \rightarrow \mathbb{R}_{>0}$ is an extensive scalar coherence character if it satisfies:

1. normalization: $\chi_n(I_n) = 1$;
2. unitary invariance: $\chi_n(UAU^*) = \chi_n(A)$;

3. direct-sum extensivity: $\chi_{m+n}(A \oplus B) = \chi_m(A)\chi_n(B)$;
4. multiplicativity on commuting positive operators;
5. continuity;
6. monotonicity in each positive spectral coordinate.

Theorem 5.2 (Determinant-character characterization). *Every extensive scalar coherence character has the form*

$$\boxed{\chi_n(A) = (\det A)^a}$$

for a single constant $a \geq 0$ independent of n . Strict clock monotonicity gives $a > 0$.

Proof. Unitary invariance reduces χ_n to a symmetric spectral function. Direct-sum extensivity reduces its logarithm to a sum of identical one-variable contributions:

$$\log \chi_n(\text{diag}(\lambda_1, \dots, \lambda_n)) = \sum_{i=1}^n g(\lambda_i).$$

Multiplicativity on one-dimensional positive scalars gives

$$g(\lambda\mu) = g(\lambda) + g(\mu).$$

Continuity yields $g(\lambda) = a \log \lambda$. Therefore $\log \chi_n(A) = a \log \det A$. Monotonicity fixes the sign. $\square$

Corollary 5.3 (Extensive scalar volume clock). *For $C_t = e^{-tK}$,*

$$\tau_{\det}(t) := -\log \chi(C_t) = -a \log \det C_t = at \operatorname{tr} K = a \operatorname{tr} \mathcal{T}_t.$$

Corollary 5.4 (Uniqueness up to unit). *Inside the extensive-character class, scalar clocks are unique up to the positive constant a .*

Non-claim 5.5 (Other scalar semantics remain). Modal clocks $\langle v, \mathcal{T}_t v \rangle$, extremal clocks $\|\mathcal{T}_t\|$, and intensive clocks $r^{-1} \operatorname{tr} \mathcal{T}_t$ answer different operational questions. They are not extensive direct-sum characters and are not ruled out as task-dependent scalarizations.

6 Universal operational clock factorization

Definition 6.1 (Clock chart). A clock chart on an observable-history domain U_i is a nondegenerate reading map

$$R_i : U_i \rightarrow \mathbb{R}.$$

Two charts are compatible when on overlaps

$$R_j = a_{ji} R_i + b_{ji}, \quad a_{ji} > 0.$$

Axiom 6.2 (Clock atlas compatibility). On every nonempty overlap the transition parameters satisfy

$$a_{ii} = 1, \quad b_{ii} = 0,$$

and on triple overlaps

$$a_{ki} = a_{kj} a_{ji}, \quad b_{ki} = a_{kj} b_{ji} + b_{kj}.$$

Transitions are independent of the path or history used to compare the same overlap.

Theorem 6.3 (Universal affine clock factorization). *Let $\{R_i : U_i \rightarrow \mathbb{R}\}$ be a connected family of clock charts with positive affine transitions satisfying identity and cocycle laws. Then there exists a one-dimensional affine clock object $\mathbb{T}$ and chart isomorphisms φ_i such that*

$$R_i = \varphi_i \circ \tau$$

for a common clock coordinate τ on the glued object. The factorization is unique up to one global positive affine transformation

$$\tau' = a\tau + b, \quad a > 0.$$

For elapsed durations, uniqueness reduces to positive scale.

Proof. Choose one chart as a gauge. Use the transition maps to transport its affine coordinate to every chart along the connected overlap graph. Cocycle closure makes the transported coordinate independent of the chosen overlap path. Any second construction differs on the reference chart by a positive affine map, and compatibility propagates the same map to all charts. $\square$

Corollary 6.4 (Clock-mechanism universality). *Independent clock mechanisms represent one operational time exactly when their calibrated readings form a positive affine atlas with path-independent cocycle-compatible transitions.*

No-go theorem 6.5 (Nonlinear drift obstruction). *If comparison of two clock mechanisms requires a transition that is nonlinear or history-dependent on an overlap, then they do not define charts of the same universal affine clock under the declared calibration class.*

Proof. A nonlinear or history-dependent transition is not an automorphism of the one-dimensional affine structure and cannot satisfy the declared path-independent affine factorization. $\square$

7 From coherence duration to physical proper time

Definition 7.1 (Physical realization ladder). The following claims are distinct:

1. a coherence process has operator duration $\mathcal{T}$;
2. a scalar clock is selected from $\mathcal{T}$;
3. multiple mechanisms factor through a universal affine clock;
4. the universal clock agrees with physical path durations;
5. path durations reconstruct a Lorentzian event geometry.

Assumption 7.2 (Relativistic realization conditions). On a declared physical domain suppose:

1. independent clock mechanisms form a universal affine clock;
2. clock differences are stable on held-out paths;
3. a directed causal relation is independently observed;
4. local event charts and interval data have sufficient rank;
5. one nondegenerate Lorentzian tensor field explains all clock-path integrals.

Theorem 7.3 (Conditional proper-time realization). *Under the relativistic realization conditions, the universal clock is identified, up to one positive unit and origin convention, with the proper-time functional of the reconstructed observable Lorentzian geometry on the tested timelike paths.*

Proof. The universal-clock theorem fixes one affine duration object. The metric-reconstruction assumption supplies a unique local quadratic form within the declared model class and full-rank design. Agreement of all calibrated clock paths with its timelike line integral identifies the affine duration with proper time up to global unit and origin. $\square$

Non-claim 7.4. This theorem is conditional. RG coherence duration has not thereby been empirically shown to equal relativistic proper time. That requires real multi-clock, multi-path, held-out experimental validation.

8 No-go theorems for time orientation and metric recovery

No-go theorem 8.1 (Unsigned duration does not determine time orientation). *A proper-time magnitude or quadratic interval cannot distinguish a displacement v from $-v$. Therefore unsigned duration data do not select future versus past.*

Proof. For any quadratic form g ,

$$g(-v, -v) = g(v, v).$$

Any scalar depending only on this value is invariant under reversal. A future cone or directed reachability relation is additional structure. $\square$

Corollary 8.2. *Semigroup direction, thermodynamic arrow, and causal time orientation are distinct until an explicit bridge is proved.*

No-go theorem 8.3 (Rank-deficient interval data do not identify the metric). *Let symmetric metric coefficients be represented by a vector $g \in \mathbb{R}^{d(d+1)/2}$ and interval observations by $y = Dg$. If D lacks full column rank, the metric is not uniquely identifiable.*

Proof. If $\text{rank } D < d(d+1)/2$, then $\ker D$ contains a nonzero h . Both g and $g + h$ yield the same interval observations. $\square$

Corollary 8.4 (Conditional local metric identification). *In a declared d -coordinate chart with a local quadratic interval model, full column rank of D identifies the symmetric metric coefficients. Signature is stable only when reconstructed eigenvalues remain separated from zero by more than uncertainty.*

9 Spacetime dimensionality and latent non-identifiability

Theorem 9.1 (Hidden-extension invariance). *Let a latent model M induce an experiment $\mathcal{E}$. For any auxiliary space Z , let $\tilde{M} = M \times Z$ and suppose all declared observation laws factor through the projection $\pi : \tilde{M} \rightarrow M$. Then M and $\tilde{M}$ induce isomorphic observational quotients and identical natural RG structures.*

Proof. Equality of observation laws is constant along each Z -fibre. Quotienting collapses those fibres, leaving the same equality-of-law classes as in M . Every RG construction natural in the experiment therefore agrees up to the induced quotient isomorphism. $\square$

Corollary 9.2 (Observable rank is not latent dimension). *Even if the regular observable quotient has dimension four and carries Lorentzian signature, the latent realization need not have dimension four.*

No-go theorem 9.3 (Observation functor need not be faithful). *Distinct latent dynamical realizations can induce the same complete observable path law. Hence latent state multiplicity, dimension, and presentation type are not experiment-internal invariants unless the declared observation functor is faithful on the competitor class.*

Proof. State-splitting provides a finite counterexample. Split one latent state into multiple states with identical aggregate transitions into each observable block and choose initial weights consistent with the lumping. The split and unsplit chains are not isomorphic as latent presentations but assign identical probabilities to every observable word. Therefore equality of complete observable path laws does not imply latent isomorphism. $\square$

Corollary 9.4 (Relative spacetime reconstruction). *A unique spacetime realization can be claimed only relative to a declared competitor category on which the clock-event observation functor is faithful.*

Non-claim 9.5 (No denial of observable spacetime geometry). The no-go results do not deny that experiments can support a regular four-dimensional Lorentzian observable quotient. They deny only the unsupported inference from that quotient to a unique latent dimension, smoothness class, topology, or manifold ontology.

10 Synthesis with fixed-protocol Resolution Geometry

Theorem 10.1 (Coherence-time extension of fibrewise RG). *Let $(\mathcal{E}, \Gamma)$ satisfy the fixed-protocol RG characterization hypotheses, and let a strictly positive observable coherence semigroup act naturally on the maximal admissible sector. Then the augmented structure*

$$\text{RG}_{\text{time}}(\mathcal{E}, \Gamma) = (Q_{\mathcal{E}}, g_F, P_{\tau}, W_{\text{obs|adm}}, \{T_{ij}\}, C_t, \mathcal{T}_t)$$

is determined fibrewise up to protocol-preserving natural isomorphism. If an extensive scalar character is required, its scalar clock is a $\text{tr } \mathcal{T}_t$. If compatible physical clock charts are supplied, their common clock is unique up to global positive affine gauge.

Proof. The quotient, metric module, projector, maximal admissible sector, and transport are fixed by the earlier fibrewise theorem. Naturality of the semigroup determines transported copies of C_t , and functional calculus determines $\mathcal{T}_t$. The determinant-character theorem handles extensive scalarization, and the universal-clock theorem handles compatible clock mechanisms. $\square$

Corollary 10.2 (Canonical hierarchy). *The mathematically justified order is*

$$(\mathcal{E}, \Gamma) \rightarrow Q_{\mathcal{E}} \rightarrow \text{observable geometry} \rightarrow C \rightarrow \mathcal{T} = -\log C \rightarrow \\ \rightarrow \text{scalar character} \rightarrow \text{universal clock} \rightarrow \text{physical realization.}$$

No later layer may be silently promoted to an earlier primitive.

11 Axiom independence and attack ledger

Removed condition	Surviving ambiguity or counterexample
Strict positivity/support control	$\log C$ may be unbounded or extended-valued on extinguished modes.
Semigroup composition	$\mathcal{T}$ need not be additive; an operational memory correction survives.
Commutation in ordered products	BCH commutator defects survive.
Direct-sum extensivity	normalized determinant and dimension-dependent scalarizations survive.
Conjugation invariance	coordinate-dependent weighted spectral clocks survive.

Continuity	pathological Cauchy-equation characters survive.
Affine transition class	arbitrary monotone recalibrations prevent a unique affine clock object.
Cocycle closure	clock calibration depends on the overlap path.
Directed causal data	future and past remain indistinguishable.
Full-rank interval design	multiple metrics fit the same observations.
Faithful competitor category	distinct latent realizations share one complete observable path law.

12 Non-claims and falsification criteria

Non-claim 12.1 (No claim that coherence is physical time). The canonical object is an operational duration of a declared coherence process. Physical time is a realization claim.

Non-claim 12.2 (No universal preference for the determinant clock). The determinant theorem applies to extensive multiplicative characters. Other operational tasks may select modal, intensive, or extremal clocks.

Non-claim 12.3 (No universal memory equals commutator claim). BCH defects quantify noncommutative logarithmic composition. Operational memory must be diagnosed from the declared multitime experiment.

Non-claim 12.4 (No protocol-free spacetime ontology). Observable Lorentzian geometry does not uniquely identify latent dimension or realization type.

Falsification criteria. The present framework requires revision if:

1. a strictly positive semigroup does not admit the stated generator/logarithm relation;
2. an extensive continuous conjugation-invariant multiplicative character exists outside the determinant-power family;
3. cocycle-compatible affine clock charts fail to admit a common affine factor;
4. a claimed universal clock requires history-dependent recalibration;
5. a claimed metric is unique despite a rank-deficient interval design;
6. a claimed latent reconstruction remains unique after an explicit same-law non-isomorphic realization is constructed;
7. physical RG clock predictions fail held-out multi-clock path comparisons.

13 Reproducibility

The theorem search was accompanied by deterministic finite batteries:

- `rg_coherence_characterization_attack.py/json`;
- `rg_clock_factorization_scalar_attack.py/json`;
- `rg_duration_character_factorization_final_attack.py/json`;
- `rg_time_dimension_attack.py/json`.

The attack seed is 20260806. The computations are not substitutes for the proofs. They test implementation, exhibit independence witnesses, and delimit stronger false claims.

14 Open frontier

Open problem 14.1 (Process-tensor extension). Replace the one-parameter semigroup by a completely positive multitime process tensor or quantum comb. Define the correct additive or cocycle-valued duration under link-product composition, and characterize the zero-memory stratum operationally. The logarithm of a channel or process tensor is not merely the logarithm of a positive state-space operator, so a separate generator space and domain theory are required.

Open problem 14.2 (Quantum measurement module). Construct the quantum experiment quotient from preparation and POVM outcome laws, select or parameterize the Petz monotone metric module, and determine which clock/asymmetry tasks select a particular metric and scalar duration.

Open problem 14.3 (Relativistic experimental realization). Calibrate an RG clock once and test held-out velocity and gravitational trajectories across independent clock mechanisms. Require a common affine clock, directed causal compatibility, and one reconstructed Lorentzian metric explaining all path durations.

15 Conclusion

Resolution Geometry does not need spacetime in order to formulate observable duration. A declared experiment first supplies an observational quotient and its admissible geometry. A strictly positive coherence semigroup then supplies the canonical operator duration $\mathcal{T} = -\log C$. Ordered noncommuting composition carries a BCH defect, while operational memory is a distinct multitime factorization question. Extensive scalar characters are determinant powers and yield the volume clock $a \operatorname{tr} \mathcal{T}$. Independent clock mechanisms define a universal operational time exactly when their positive affine charts satisfy path-independent cocycle gluing.

Physical proper time and spacetime occur only at later realization layers. Unsigned duration does not select time orientation; insufficient interval data do not identify a metric; observable Lorentzian rank does not identify latent dimension; and the observation functor may fail to distinguish non-isomorphic latent dynamics. The strongest defensible conclusion is therefore both positive and limited:

RG canonically characterizes observable duration and universal clock structure, while physical time and spacetime remain experimentally constrained realizations.

References

- [1] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, 1966.
- [2] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS and Oxford University Press, 2000.
- [3] D. Blackwell, “Comparison of experiments,” *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, pp. 93–102, 1951.
- [4] D. Blackwell, “Equivalent comparisons of experiments,” *Annals of Mathematical Statistics*, vol. 24, pp. 265–272, 1953.
- [5] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, AMS, 1982.
- [6] K.-J. Engel and R. Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Springer, 2000.

- [7] B. C. Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed., Springer, 2015.
- [8] M. Haase, “The functional calculus approach to the spectral theorem,” *Indagationes Mathematicae*, vol. 33, no. 1, pp. 17–47, 2022.
- [9] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Communications in Mathematical Physics*, vol. 48, pp. 119–130, 1976.
- [10] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.
- [11] D. Petz, “Monotone metrics on matrix spaces,” *Linear Algebra and its Applications*, vol. 244, pp. 81–96, 1996.
- [12] V. S. Varadarajan, *Lie Groups, Lie Algebras, and Their Representations*, Springer, 1984.
- [13] M. Rada, “Resolution Geometry of Observable Systems: Observable Quotients, Admissible Distinguishability, Transport, and Entropy Decomposition,” preprint, 2026.
- [14] M. Rada, “Fibrewise Characterization of Resolution Geometry,” preprint, 2026.

A Finite attack summary

The associated batteries verified the following finite statements:

1. $-\log(e^{-tK}) = tK$ on strictly positive sampled modes;
2. logarithmic duration is additive for multiplicative scalar survival;
3. inserted scalar memory corrections are recovered as additivity defects;
4. consistent memory assignments satisfy the tested cocycle identity;
5. determinant powers satisfy the extensive-character axioms;
6. trace mean and extremal-eigenvalue maps fail those axioms;
7. compatible affine clock charts satisfy cocycle closure and reconstruct a common clock;
8. nonlinear path-dependent recalibration fails affine factorization;
9. full-rank metric designs identify all symmetric coefficients while degenerate designs do not;
10. hidden extensions and state-split latent chains preserve the declared observable laws.

B Notation ledger

Symbol	Meaning
$\mathcal{E}$	complete declared statistical experiment
Γ	admissibility protocol
$Q_{\mathcal{E}}$	observational quotient
C_t	observable coherence process
K	positive coherence-loss generator
$\mathcal{T}_t$	operator-valued duration $-\log C_t$
$\mathcal{M}^{\text{BCH}}$	noncommutative logarithmic composition defect
$\mathcal{M}^{\text{op}}$	operational multitime memory obstruction

χ	scalar coherence character
τ_{det}	extensive determinant/volume clock
R_i	reading in clock chart i
$\mathbb{T}$	universal affine clock object
